

Soil moisture relationship

- ① Gravitational water :- $\rightarrow$ not for plant use
- not held by soil, but drain out freely by gravity action hence not available for plant use.

- ② Capillary water :- $\rightarrow$ available water
- $\rightarrow$ held by surface tension

- ③ Hygroscopic water :- $\rightarrow$ not for plant use
- $\rightarrow$ not removed easily
- $\rightarrow$ Below PWP soil has hygroscopic water

- ④ Saturation capacity :- max. water holding capacity of soil, when all pores filled with water
- Soil moisture tension = 0

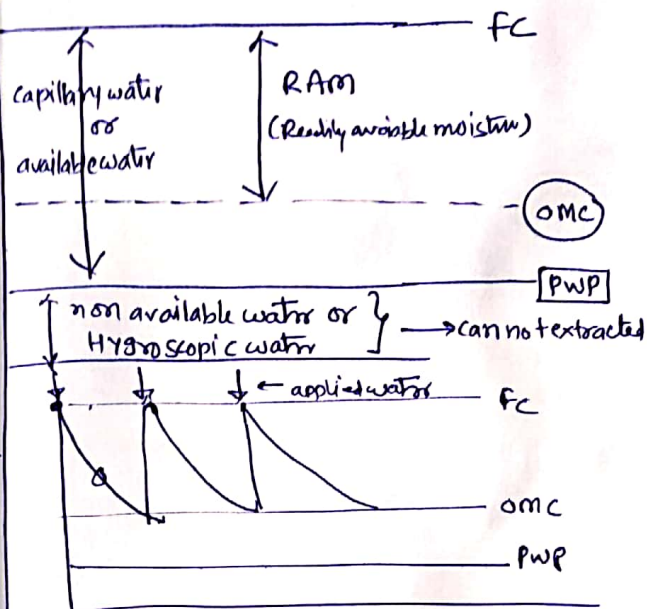
- ⑤ FC (field capacity) Soil moisture $\Rightarrow \frac{1}{3} - \frac{1}{10}$ atm tension
- $\rightarrow$ upper limit of capillary water
 - $\rightarrow$ after gravity water drain off, max. amount of water which can be held up by soil against gravity.

Depth of water stored in root zone (in filling the soil upto FC)

$$dw = d \frac{\gamma_d}{\gamma_w} (FC) = m_d$$

$$\therefore m = \gamma_d / \gamma_w * FC$$

- ⑥ PWP Soil moisture tension $\Rightarrow 7-32$ atm.



Type-1 :- $FC = 28\%$ $PWP = 15\%$
 $RAM = 80\%$ of AM

Soln :- $FC = 28\%$
 $RAM = 80 \times \frac{13}{100} = 10.4\%$
 $OMC = 12.6\%$
 $PWP = 15\%$

$AM = 28 - 15 = 13\%$
 $\therefore OMC = 28 - 10.4 = 17.6\%$

$\therefore dw = d \frac{\gamma_d}{\gamma_w} [28 - 17.6]$
 or (last 4) $\rightarrow RAM$

Type-2 :- $FC = 40\%$ $PWP = 18\%$

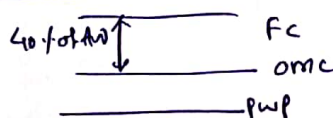
Mc falls to 40% of Available water
 or Mc falls by 60% of Available water

Soln :- $RAM = \frac{60}{100} \times (40 - 18) = 13.2$

$FC = 40$
 $13.2 = RAM$
 $OMC = 26.8$
 $PWP = 18$

$d \frac{\gamma_d}{\gamma_w} (40 - 26.8) = d$ (13.2)

Type-3 :- Allowable depletion = 40% of AM



note :- $\frac{1}{3}$ rd atm. moisture point $\Rightarrow$ % of moisture retain in soil sample when placed on porous plate & subjected to Pressure = atm pressure / 3